MILITARY SPECIFICATION

INDICATOR, TEMPERATURE, ELECTRICAL RESISTANCE, MULTI-FUNCTION, -70° TO +150°C

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers multi-function temperature indicators with a range from -70° to +150° Centigrade (C).

1.2 Classification. The indicators shall be furnished in the following part numbers, as specified (see 6.2).

MS28009-2 Fluorescent Dial
MS28009-4 Lusterless White Dial

2. APPLICABLE DOCUMENTS

* 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636	Box, Fiberboard

Military

MIL-P-116	Preservation, Methods Of
MIL-I-7070	Indicator, Temperature, Electrical Resistance, Hermetically Sealed, General Specification For
MIL-L-25142	Luminescent Material, Fluorescent

STANDARDS

Federal

FED-STD-595	Colors

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Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MS28009	Indicator - Temperature, Multi-Function, Electrical, Resistance, Single, Clamp-Mounted, -70 to 150 Degrees C, 2-Inch Size
MS33585	Pointers - Dial, Standard Design Of Aircraft Instruments

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. The indicator furnished under this specification shall be a product which has been tested and has passed the qualification tests specified, and has been listed on or approved for listing on the applicable Qualified Products Lists (QPL).

3.2 General: The requirements specified in MIL-I-7070 are applicable as requirements of this specification: Additional requirements shall be as specified herein:

3.2.1 Precedence: In the event the requirements of the general specification and this specification conflict; the requirements of this specification shall govern.

3.3 Construction.

3.3.1 Case: The indicator case shall conform to MS28009:

3.3.2 Dial. The dial shall conform to Figure 1. The diameter of the dial as measured across the outer ends of the graduations shall not be less than 1-3/4 inches.

3.3.2.1 Scale arc. The indicator shall provide a minimum scale arc angle of 250 degrees. The mechanism shall not be geared to obtain the specified scale arc:

3.3.3 Pointer. The dimensions of the pointer tip shall conform to MS33585-8:

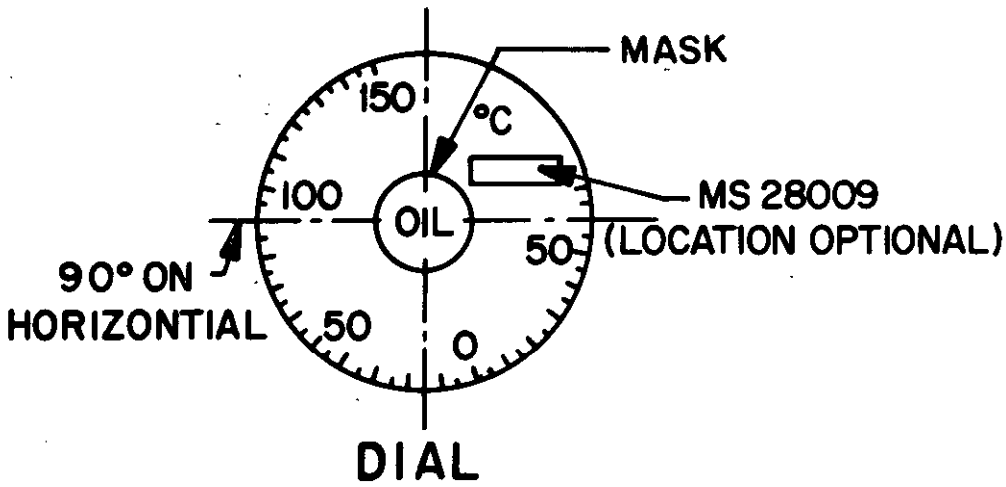
3.3.3.1 Pointer oversving: The pointer oversving shall be not less than 3 angular degrees beyond each end of the scale.

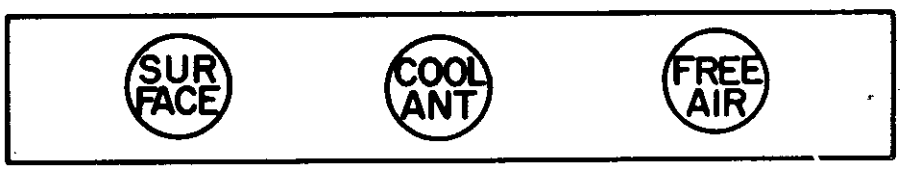
3.4 Markings:

3.4.1 Dial markings. The markings shall be durable and in accordance with Figure 1. The dimensions of the dial markings and the type of paint used shall be as shown in Table 1. The fluorescent-luminescent shall conform to MIL-L-[illegible]. The lusterless white shall be Color 37875 of FED-STD-595. The lusterless black shall be Color 37038 of FED-STD-595.

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IDENTIFICATION DECALCOMANIA

FIGURE 1

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3.4.2 Mask. The mask shall be either a disk of suitable metal, or a decalcomania affixed firmly and permanently to the center of the inside of the cover glass and shall contain the word "Oil."

3.5 Weight. The weight of the complete indicator shall not exceed 1 pound.

3.6 Electrical receptable. An electrical receptacle shall be provided and located as shown on MS28009. The receptacle shall be wired as follows:

Terminal	Connection
A	+28 Volt Battery
B	Resistance Bulb
C	Common - Negative 24 Volt Battery and remaining resistance bulb connection
D	Open

3.7 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

TABLE I - Dial Markings

Marking	Height or Length Inch ±0.005	Width of Line or Graduation Inch ±0.005	Finish
Numerals	0.125	0.020	
50°C Graduations	0.156	0.020	Fluorescent
10°C Graduations	0.094	0.015	Luminescent
5°C Graduations	0.063	0.010	or
Lettering "°C"	0.125	0.020	Lusterless
Mask Lettering	0.125	0.020	White
Decalcomania Lettering	0.125	0.020	
Visible Portion of Pointer	.....	.....	
MS Part Number	0.062	0.010	Lusterless
Background of Dial and Mask	.....	.....	Black

3.8 Instrument identification decalcomania. The indicators "MS28009-2," and "MS28009-4" shall be provided with identification decalcomanias as shown on Figure 1. The decalcomania shall be larger in diameter than the mask, but not of a size to interfere with readability of the indicator. The decalcomania shall be of the water-applied type and shall be suitable for application to the cover glass of the indicator. The decalcomania, when mounted on the cover glass of the indicator, shall be capable of meeting the tests specified in this specification, without peeling, blistering, fading, loss of adhesion, checking softening, or hardening.

3.8.1 Each decalcomania shall be packaged in an envelope marked: "THIS ENVELOPE CONTAINS IDENTIFICATION DECALCOMANIA." Complete and detailed instructions for mounting the decalcomania on the cover glass of the indicator shall be furnished with each decalcomania.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General specification. The indicator shall be subjected to the Quality Assurance Provisions in accordance with MIL-I-7070 except as specified herein.

4.3 Test conditions.

4.3.1 Test voltage. The test voltage shall be 28.5 ±0.5 volts direct current (DC), unless otherwise specified.

4.3.2 Test tolerances. The tolerances specified in Table II, not inclosed in parentheses, are applicable only when the indicators are subjected to all the tests in the sequence specified herein. The tolerances specified in Table II, inclosed in parentheses, are applicable only when the indicators are subjected to a specific test independent of sequence.

4.4 Test methods.

4.4.1 Individual tests.

4.4.1.1 Scale error at room temperature. The test resistance specified in Table II shall be applied to the resistance bulb terminals of the indicator. The test voltage shall be applied and the scale errors at the test points specified in Table II shall not exceed the tolerance specified in Table II.

4.4.1.2 Position error. The position errors of the indicator shall not exceed 3°C.

4.4.1.3 Friction error. The friction errors of the indicator shall not exceed 3°C.

4.4.1.4 Sealing. No leaks shall result from the sealing test.

4.4.1.5 Insulation leakage. The insulation resistance shall be a minimum of 20 megohms.

* 4.4.1.6 Fogging. The indicator shall be placed in a 71° ±2°C controlled ambient temperature for a minimum of 1 hour. At the end of this period and while still at this temperature, an ice cube shall be rubbed on the indicator glass face for 1 minute ±5 seconds. The glass shall be dried (do not use compressed air). Evidence of water or oil condensate shall be cause for rejection.

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TABLE II - Test Tolerances

Test Point	Test Resistance	Scale Error at Room Temperature	Reduce Voltage	Scale Error at Low Temperature	Scale Error at High Temperature	Scale Error at Room Temperature After Vibration
°C	Ohms	Tol ±°C	Tol ±°C	Tol ±°C	Tol ±°C	Tol ±°C
-70	68.27	4.0 (5.0)	3.0 (3.0)	6.5 (6.5)	6.5 (6.5)	6.5 (5.5)
-30	80.56	3.0 (3.0)	3.0 (3.0)	5.5 (5.5)	5.5 (5.5)	6.5 (5.5)
0	98.38	2.0 (2.0)	2.0 (2.0)	4.0 (4.0)	4.0 (4.0)	5.5 (5.0)
+30	100.91	2.0 (2.0)	2.0 (2.0)	4.0 (4.0)	4.0 (4.0)	5.5 (5.0)
80	120.36	2.0 (2.0)	2.0 (2.0)	4.0 (4.0)	4.0 (4.0)	5.5 (5.0)
120	137.78	3.0 (3.0)	3.0 (3.0)	5.0 (5.5)	5.5 (5.5)	6.5 (6.5)
150	151.91	4.0 (5.0)	3.0 (3.0)	6.0 (6.5)	6.5 (6.5)	6.5 (6.5)

Test Point	Test Resistance	Scale Error at Room Temperature After Open And Short Circuit	Scale Error With Bulb Lead Resistance	Extreme Temperature Exposure		
				Scale Error At Low Temperature	Scale Error At Room Temperature After Low Temperature	Scale Error At Room Temperature After High Temperature
°C	Ohms	Tol ±°C	Tol ±°C	Tol ±°C	Tol ±°C	Tol ±°C
-70	68.27	6.5 (5.0)	3.0 (3)	9.5 (6.5)	6.5 (4.0)	6.5 (4.0)
-30	80.56	6.5 (4.0)	3.0 (3)	9.5 (5.5)	6.5 (3.0)	6.5 (3.0)
0	98.38	5.5 (3.0)	3.0 (3)	7.5 (4.0)	5.5 (2.0)	5.5 (2.0)
+30	100.91	5.5 (3.0)	3.0 (3)	7.5 (4.0)	5.5 (2.0)	5.5 (2.0)
80	120.36	5.5 (3.0)	3.0 (3)	7.5 (4.0)	5.5 (2.0)	5.5 (2.0)
120	137.78	6.5 (4.0)	3.0 (3)	9.5 (6.5)	6.5 (3.0)	6.5 (3.0)
150	151.91	6.5 (5.0)	3.0 (3)	9.5 (6.5)	6.5 (4.0)	6.5 (4.0)

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4.4.2 Sampling plan A tests.

4.4.2.1 Reduced voltage operation. The indicator shall be subjected to a scale error at room temperature test with an applied voltage of 22.5 volts at the test points specified in Table II. The indications at the reduced voltage shall not differ from the indications with 28.5 volts applied by more than the tolerances specified in Table II.

4.4.2.2 Scale error at low temperature. The scale error shall not exceed the tolerances specified in Table II.

4.4.2.3 Scale error at high temperature. The scale error shall not exceed the tolerances specified in Table II.

4.4.2.4 Magnetic effect. The magnetic compass shall not deflect more than 5 degrees.

4.4.2.5 Resistance bulb current. A total resistance of 100 ohms shall be applied across the resistance bulb terminals of the indicator, the test voltage applied, and the resistance bulb circuit current measured. The current shall not exceed 22 milliamperes.

4.4.2.6 Vibration error. The pointer oscillation shall not exceed 2°C. The pointer variation shall not exceed 2°C.

4.4.2.7 Vibration failure. The scale errors after vibration shall not exceed the tolerances specified in Table II.

4.4.2.8 Open and short circuit. The scale errors after open and short circuit shall not exceed the tolerances specified in Table II.

4.4.3 Sampling plan B tests.

4.4.3.1 Bulb lead resistance effect. The indicator shall be connected as shown on Figure 2 and subjected to a scale error at room temperature test. The indications with bulb lead resistance shall not differ from the indications without bulb lead resistance, by more than the tolerances specified in Table II.

4.4.3.2 Response time. The response time shall not exceed 3 seconds.

4.4.3.3 Damping factor. The damping factor shall not be less than 1.6.

4.4.3.4 Power consumption. The power consumption shall not exceed 2 watts.

4.4.3.5 Extreme low temperature exposure. The scale errors at low temperature shall not exceed the tolerances specified in Table II. The scale errors at room temperature after low temperature exposure shall not exceed the tolerances specified in Table II.

4.4.3.6 High temperature exposure. The scale errors at room temperature after high temperature exposure shall not exceed the tolerances specified in Table II.

4.4.3.7 Thermal shock. No damage to the indicator shall result from the thermal shock test.

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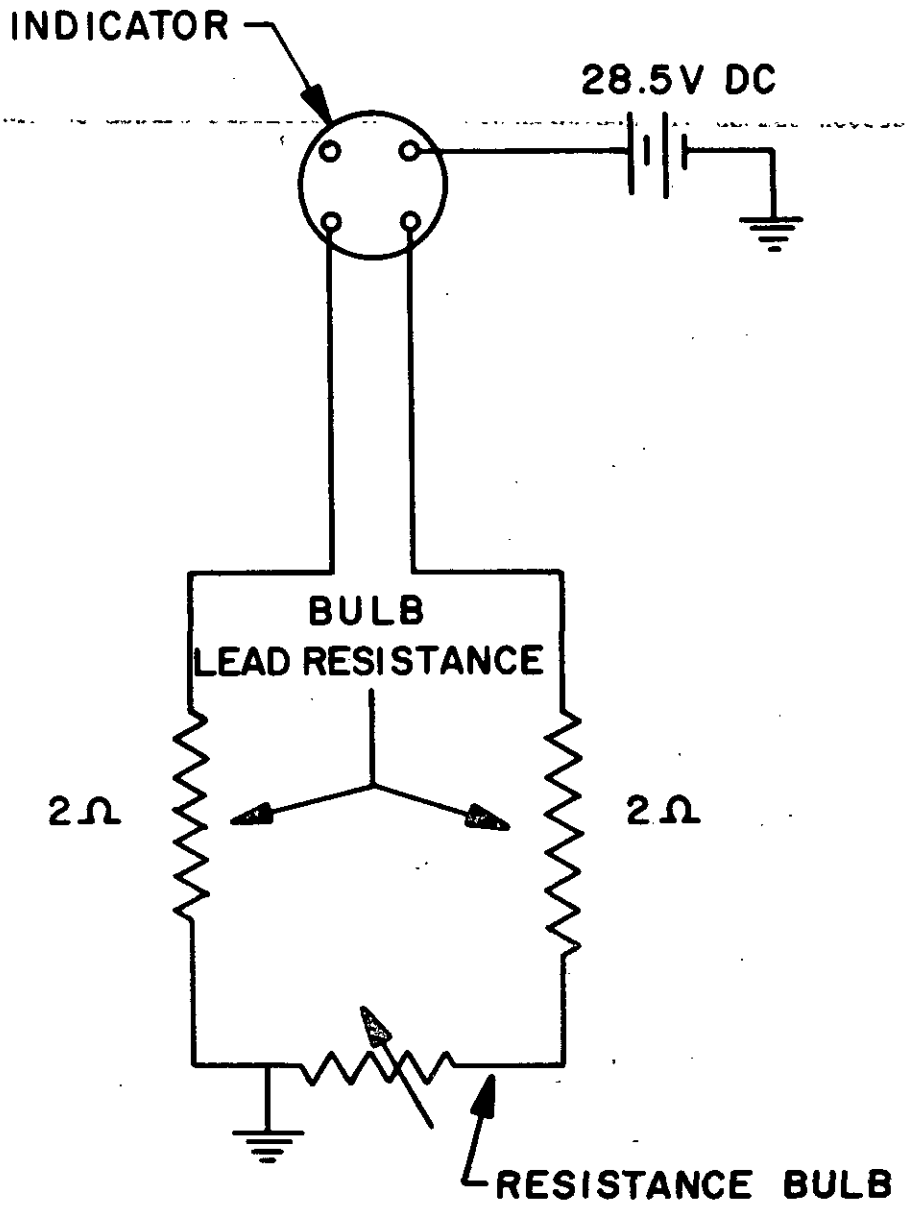


FIGURE 2
WIRING DIAGRAM

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* 4.4.3.8 Salt spray. The indicator shall be subjected to and shall meet the requirements of the sealing test following salt spray exposure. The indicator shall be subjected to a scale error test after salt spray. The insulation resistance shall be a minimum of 20 megohms.

5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified (see 6.3).

* 5.1.1 Level A. Each indicator shall be preserved and packaged in accordance with Method III of MIL-P-116 and placed in a snug fitting container conforming to PPP-B-636, Grade W6C.

* 5.1.2 Level C. Each indicator shall be preserved and packaged in a manner which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity. The supplier may use his commercial practice providing it fulfills this requirement.

* 5.2 Packing. Packing shall be Level A, B or C as specified (see 6.3).

* 5.2.1 Level A. Indicators preserved and packaged as specified in 5.1.1 shall be packed in weather-resistant shipping containers conforming to PPP-B-636. The grade of each shipping container, determined by the gross weight and size, shall be limited to the applicable special requirements table of the specification. Where practicable, the exterior shipping container shall be of uniform shape and size and of minimum cube and tare. Closure and strapping shall be in accordance with the appendix of PPP-B-636.

* 5.2.2 Level B. Same as Level A except class domestic shipping containers may be used.

* 5.2.3 Level C. Packages that require overpacking for acceptance by the carrier shall be placed in exterior type shipping containers in a manner that will insure safe transporation at the lowest rate to the point of delivery. Containers shall comply with Uniform Freight Classification Rules or regulations of other carriers as applicable to the mode of transportation.

* 5.3 Marking. The interior and exterior shipping containers shall be marked in accordance with MIL-STD-129. The nomenclature shall be as follows:

(FSN) Indicator, Temperature, Electrical Resistance, Multi-Function, -70° to +150°C, 2 inch size.

6. NOTES

6.1 Intended use. The electrical resistance temperature indicator covered by this specification is intended to indicate temperature in the range of -70° to +150°C, such as oil, coolant, surface and free air.

6.2 Bezel. The temperature indicator covered by this specification can be used on instrument panels requiring MS33038 case mounting by the application of an instrument mounting bezel in accordance with [illegible].

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* 6.3 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of specification.

b. Designate indicator desired (see 1.2).

c. Ordering data in accordance with MIL-I-7070.

d. Level of packaging and packing required (see 5.1 and 5.2).

* 6.4 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the San Antonio Air Materiel Area, Service Engineering Division, ATTN: MMEE, Kelly AFB, Texas 78241, and information pertaining to qualification of products may be obtained from that activity.

* 6.5 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Air Force - 82
Navy - AS

Review Activity:
Navy - AS

Preparing Activity:
Air Force - 82

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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